



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO CSK FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

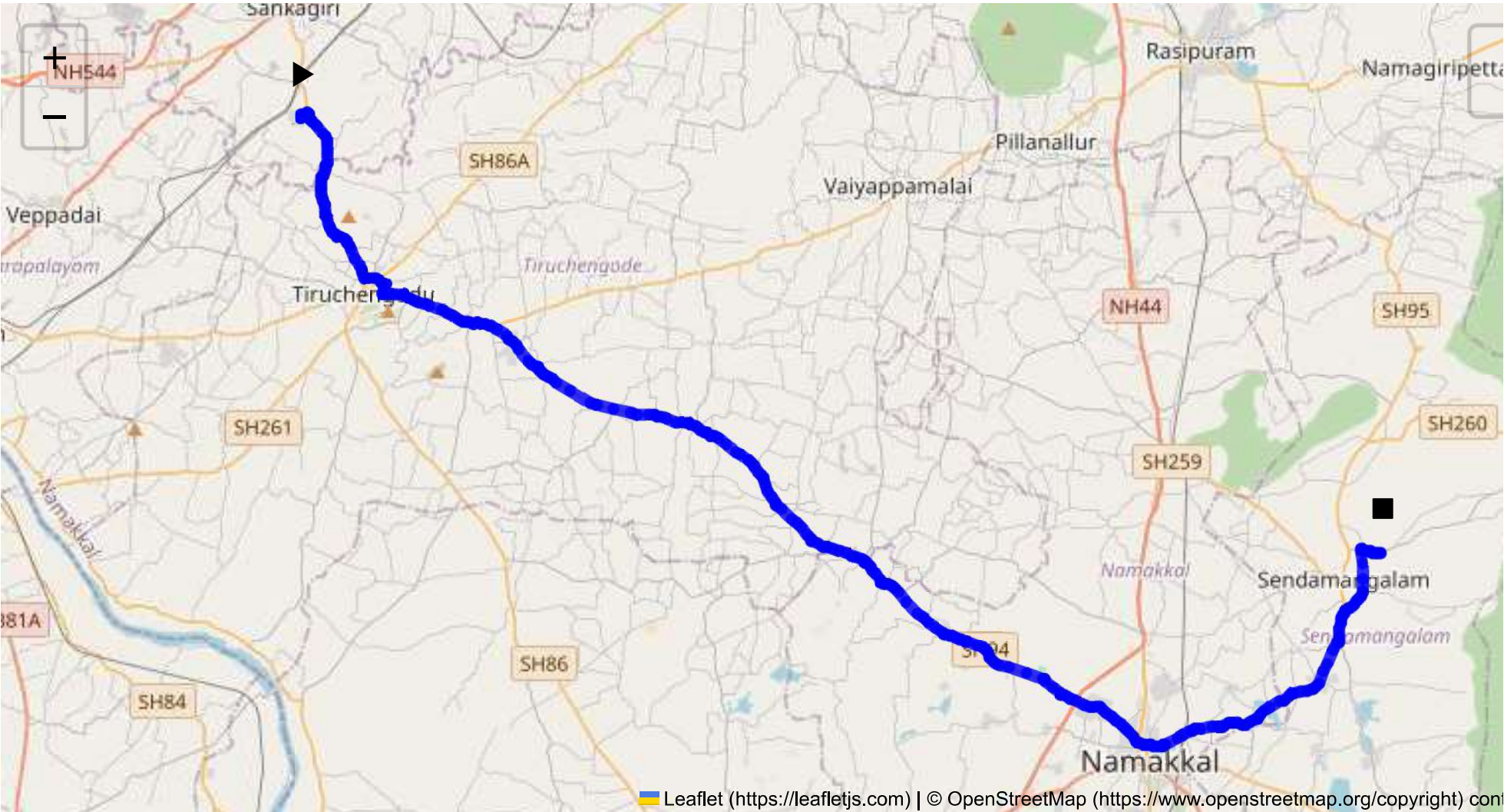
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 57.38 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.290347, 78.248012)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Periyakulam is approximately 57.38 kilometers, primarily following regional roads that interconnect these towns in Tamil Nadu. The route traverses through rural and semi-urban landscapes with a mix of straight and winding sections.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences a tropical climate with distinct dry and wet seasons. The route may encounter:

- **Monsoon Season (June to September):** Risks of heavy rainfall leading to slippery roads and reduced visibility.
- **Summer Season (March to May):** High temperatures might affect vehicle cooling systems, increasing the risk of mechanical failures.
- **Cyclones:** Rare but can cause severe weather disruptions, although more prevalent along the coastal areas.

3. Traffic Patterns

- **Peak Hours:** Typically from 8 AM to 10 AM and 5 PM to 8 PM due to local commuter traffic.
- **Congestion-prone Areas:** Town centers and entry/exit points of Sangagiri and Periyakulam may experience slow-moving traffic.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Generally maintained, but some segments may have potholes, especially after monsoon rains. Road signage and lighting may be inadequate in rural areas.
- **Infrastructure:** Few bridges and overpasses; expect level crossings.

5. Suggestions for Alternative Routes for Emergencies

- The alternative would involve local roads that run parallel to the primary route, though they might have lesser quality and increased travel time. Using mobile navigation apps can provide real-time detour options.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Required for transport of hazardous materials; adherence to specific timings and routes is crucial.
- **Safety Regulations:** Compliance with Tamil Nadu Environmental Safety norms and driver certification for hazardous materials.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There are no major historical incidents recorded specifically for this route, but common issues include overturning due to sharp turns, particularly on smaller roads.

8. Environmental Considerations and Sensitive Areas

- **Reserve forests:** Any nearby reserved forest areas might need special transit permissions.
- **Water bodies:** Rivers or water tanks in proximity require careful navigation to prevent contamination during an incident.

9. Analysis of Communication Coverage

- Adequate coverage in most areas due to cellular networks improving; however, some rural stretches might experience intermittent connectivity.

10. Estimated Emergency Response Times

- **Urban Areas:** Typically 30 to 45 minutes.
- **Rural Areas:** Can extend to over an hour due to distance and road conditions from emergency services based in larger towns.

12. Overall Summary of Risk Assessment

The route poses moderate risk levels for the transportation of hazardous materials. Key challenges include weather impacts, road quality variability, and limited emergency response reach. Effective measures include careful monitoring of weather forecasts, routine vehicle maintenance checks, and adherence to regulatory requirements. Communication devices should be checked regularly, and alternative routes should be known in advance to manage unforeseen disruptions efficiently.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	Medium	11.38342, 77.89975	30 KM/Hr
5	Turn	Medium	11.38295, 77.90001	30 KM/Hr
6	Turn	High	11.38186, 77.90258	15 KM/Hr
7	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
8	Turn	Medium	11.22480, 78.16534	30 KM/Hr
9	Turn	Medium	11.22406, 78.17197	30 KM/Hr
10	Turn	Medium	11.24228, 78.21699	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.24477, 78.22620	30 KM/Hr
12	Blind Spot	Blind Spot	11.29162, 78.24142	10 KM/Hr
13	Turn	Medium	11.29045, 78.24643	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
16	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
17	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
18	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
19	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
20	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
21	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
22	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
23	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
24	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
28	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
29	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
31	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
32	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
33	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
34	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
35	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
36	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
37	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
38	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
39	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
40	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
25	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
26	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
30	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Coordinates: 11.38295, 77.90001



Coordinates: 11.38186, 77.90258



Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22480, 78.16534



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22406, 78.17197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24228, 78.21699



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24477, 78.22620



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.29162, 78.24142



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.29045, 78.24643